Rithvik Santosh Shetty

AI Engineer | Applied ML, Computer Vision, LLM and Voice AI Systems
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Summary

AI engineer with a B.E. in Artificial Intelligence and Machine Learning. Shipped a YOLOv8 vision pipeline into an industrial workflow, a multilingual voice AI assistant that placed Top 12 at Sarvam AI's national hackathon, and a PyTorch model with GNN encoders for cancer drug response. Python and PyTorch for models, FastAPI and Next.js around them.

Technical Skills

- **Languages:** Python, TypeScript, JavaScript, SQL
- **ML and Data:** PyTorch, scikit-learn, XGBoost, Pandas, NumPy, Jupyter, deep learning, graph neural networks, ensemble methods
- **Vision, NLP and Speech:** YOLOv8, OpenCV, Sentence-BERT, speech-to-text, text-to-speech, translation
- **LLMs and Agents:** Google Gemini API, Sarvam AI APIs, prompt engineering, agentic workflows, Model Context Protocol (MCP), RAG-style retrieval over structured data
- **Backend and Product:** FastAPI, Node.js, REST APIs, PostgreSQL, Supabase, Redis/Upstash, Cloudflare Workers, Twilio and WhatsApp APIs, React, Next.js, React Native, Streamlit, Git, Linux, Vercel

Experience

Ion Chemicals Jan 2025 – Apr 2025
Python Developer Intern *Navi Mumbai, India*

- Engineered a real-time computer vision pipeline with YOLOv8, OpenCV and Python that detects gas leaks and anomalies on industrial pipelines, reducing manual inspection time by 40%.
- Integrated the detector into existing operational workflows with the plant engineering team.

TRCT (trct.in), The Run Club Thane Mar 2024 – Present
Freelance Full-Stack Developer *Remote*

- Sole developer on the club's production Next.js and TypeScript platform for a 4,000+ member community: authentication, REST APIs, a Razorpay and Stripe e-commerce module, and a 100 Lighthouse score.

Projects

Praman | *Python, FastAPI, Sarvam AI, Twilio WhatsApp API* 2026

- Built a WhatsApp banking and loan assistant for low-literacy rural users: voice or text in and out, in the user's own language, on Sarvam AI speech-to-text, translation and text-to-speech across 22 Indian languages.
- Added a document explainer for scanned bank letters and loan agreements, plus an operator dashboard. **Top 12 of ~100 teams** (~6,000 registrations), Sarvam Buildin' Hours.
- Rebuilt it for health insurance claims at the Paytm Build for India AI Hackathon: users send claim documents and voice notes on WhatsApp in 11 Indian languages, and agents handle the synced threads from a dashboard. Code: github.com/rithvikshettyy/praman-paytm

AutoPharmaX | *Python, PyTorch, XGBoost, scikit-learn, Streamlit* Jul 2025

- Built a multitask deep learning model in PyTorch with dual MLP and graph neural network encoders to predict cancer drug response and drug-pair synergy, benchmarked against XGBoost and scikit-learn ensembles.
- Added agentic tooling to query drug-efficacy data and a chatbot that turns model outputs into plain-language guidance. Code: github.com/rithvikshettyy/autopharmax
- Wrote up the method and results as a conference paper with the team; reviewers returned a revise-and-resubmit.

AnyFormat | *Next.js, TypeScript, Upstash Redis, BullMQ, Razorpay* 2026

- Self-hosted file conversion SaaS: queued jobs over Sharp, MuPDF, FFmpeg, LibreOffice and Tesseract OCR, with auth, payments and bot protection. Live: anyformat.rithvikshetty.in

SeaGuard | *React Native, FastAPI, Twilio* Sep 2025

- Safety app for fishermen: real-time GPS boundary monitoring, weather alerts and automated SMS alerts. Code: github.com/rithvikshettyy/SeaGuard

Education

SIES Graduate School of Technology (affiliated to the University of Mumbai) Nov 2022 – May 2026
Bachelor of Engineering (B.E.) in Artificial Intelligence and Machine Learning *Navi Mumbai, India*

- Coursework: Machine Learning, Deep Learning, NLP, Image and Video Processing, Advanced AI, Big Data Analytics, Data Analytics and Visualization

Achievements

- **Top 12 of ~100 on-site teams** (~6,000 registrations), Sarvam Buildin' Hours hackathon by Sarvam AI, Bengaluru (2026)
- Selected among ~100 finalist teams from 6,000+ registrations, Paytm Build for India AI Hackathon, Mumbai (2026)
- **Most Product Readiness Award**, HackCelestial 2.0 (200+ teams), Pillai College of Engineering (2025)
- Award for SeaGuard presented by the Director, ISRO (2025)

Certifications and Activities

- Introduction to Machine Learning on AWS (AWS and Coursera, 2025) | Introduction to Model Context Protocol (Anthropic, 2025) | Secretary, Technical Team, SIES GST: co-organized ByteCamp'26, a national hackathon with 3,000+ registrations